

Pre-Read: Comparison, Logical Operators & Conditional Statements



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Pre-Reads

Foundation

Recommended

Resource



Associated Lectures (1)



Description



Discussions



Pre-Lecture Notes: Comparison, Logical Operators & Conditional Statements



1. Introduction

Imagine you are a robot that makes decisions. You check if it's raining 🌧 before going outside, or see if your marks are higher than 90 to get a reward 🎁.

That's exactly what **comparison** and *conditional statements* help Python do - **make smart decisions** based on conditions.



2. Comparison Operators

Comparison operators compare two values and tell us **True** or **False**.

Operator	Meaning	Example	Output
==	Is equal to	5 == 5	✅ True
!=	Not equal to	5 != 3	✅ True
>	Greater than	8 > 5	✅ True
<	Less than	2 < 5	✅ True
>=	Greater than or equal to	6 >= 6	✅ True
<=	Less than or equal to	3 <= 5	✅ True



Example:

```
a = 10
b = 20

print(a == b) # False
print(a < b)  # True
print(a != b) # True
```

💡 Real-life example:

Think of comparing ages:

```
my_age = 12
friend_age = 14

print(my_age > friend_age) # False, because 12 is not greater than 14
```

⚙️ 3. Logical Operators

Logical operators combine multiple conditions.

Operator	Meaning	Example	Output
and	True if both conditions are true	(5 > 3) and (8 > 6)	✅ True
or	True if any one condition is true	(5 > 3) or (8 < 6)	✅ True
not	Reverses the result	not (5 > 3)	❌ False

💡 Real-life example:

You can go out **if it's sunny AND you finished homework**:

```
is_sunny = True
homework_done = True

if is_sunny and homework_done:
    print("You can go out and play!")
```

If **either** condition works:

```
is_weekend = False
holiday = True

if is_weekend or holiday:
```

```
print("No school today!")
```



4. Conditional Statements

Conditional statements let Python **decide** what to do.

Basic if

```
age = 18
if age >= 18:
    print("You are an adult!")
```

if...else

```
age = 15
if age >= 18:
    print("You can vote!")
else:
    print("You are too young to vote.")
```

if...elif...else

```
marks = 85

if marks >= 90:
    print("Excellent!")
elif marks >= 70:
    print("Good job!")
else:
    print("Keep practicing!")
```



Real-life analogy:

If you score 90+, you get a chocolate 🍫, if 70-89, you get a toffee 🍬, otherwise, keep trying 💪.



5. Nested Conditions

Sometimes you need to check a condition *inside* another condition.

```
age = 20
has_id = True

if age >= 18:
    if has_id:
        print("You can enter the club.")
    else:
        print("You need an ID.")
else:
    print("You are too young.")
```



6. Visual Understanding

Here are some helpful visual guides:

- **Comparison Operators Explained:** <https://cdn.educba.com/academy/wp-content/uploads/2019/09/Python-Comparison-Operators-1.png>
- **Conditional Flow Chart:** <https://media.geeksforgeeks.org/wp-content/uploads/if-elseif-ladder.jpg>
- **Logical Operators (and/or/not):** https://d8it4huxumps7.cloudfront.net/bites/wp-content/banners/2024/2/65df0c8246ce1_python_logical_operators-_banner.jpg?d=1200x800



7. Common Beginner Mistakes

Mistake	Example	Correct Way
Using = instead of ==	if a = 5: ❌	if a == 5: ✅
Forgetting indentation	if True: print("Hi") ❌	 Alt text ✅
Writing elseif instead of elif	elseif ❌	elif ✅
Using quotes wrongly	if "5" > 3: ❌	if int("5") > 3: ✅



8. Summary

- Comparison operators → check relationships.
- Logical operators → combine conditions.
- Conditional statements → make decisions.
- Indentation → very important in Python!
- Think logically like: “If this, then that.”